JNTUH USED 19-12-18 AM Set

Code No: 117HN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, November/December - 2018
SOFT COMPUTING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) How to improve the effectiveness of a search based problem solving technique? [2]
b) What is a constraint satisfaction problem? [3]
c) List various models of ANN. [2]
d) What is adaptive linear neuron? Explain. [3]
e) Explain what is hamming network. [2]
f) What is 'learning vector quantization'? [3]
g) What is fuzzy logic? Explain. [2]
h) Differentiate between crisp set and fuzzy set. [3]
i) What are the disadvantages of genetic algorithms? [2]
j) State the different selection methods in GA. [3]

PART-B

(50 Marks)

2.a) What is Artificial Intelligence? Mention some of the applications that fall within the scope of AI.
b) How can you represent the resolution of predicate logic? [5+5]

OR

3.a) Explain the algorithm for steepest hill climbing.
b) Discuss about means end analysis in brief. [5+5]

4. What is supervised learning in ANN? Explain different learning law's with examples. [10]

OR

5. Explain the storage and retrieval algorithms in associative memories. [10]

6. Explain about self organizing Map and its training algorithms. [10]

OR

7. Explain the principle of Maxnet and write algorithm for Maxnet. [10]

8.a) What is union in Fuzzy set operation and intersection in Fuzzy operation? Give examples.
b) Define a Fuzzy Cartesian product and explain. [5+5]

OR

9.a) Explain the concept of fuzzification with an example.
b) Discuss about the Centre of Gravity method of defuzzification. [5+5]

10. Discuss the algorithmic steps involved in solving an optimization problem using Tabu search with suitable examples. [10]

OR

11.a) Explain the measures of fuzziness in detail.
b) What is the main function of cross over operation in Genetic Algorithm? [5+5]

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